

Delta-Neutral Grid Market Making with Adaptive Hedging

and Risk Management for Cryptocurrency Spot–Futures Markets

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Abstract

We present a systematic, delta-neutral market-making framework for cryptocurrency spot–futures pairs, exemplified on BTCUSDT on Binance. The architecture combines two-sided grid order books with a cross-venue hedging mechanism that continuously maintains near-zero directional exposure while harvesting bid–ask spread, grid profit, funding fees, and maker volume rebates. We introduce two complementary hedging logics—a static spot holding with a futures sell-only grid (Mode 1), and a fully active symmetric long-spot/short-futures dual grid (Mode 2)—and analyse their trade-offs under varying market regimes. The framework further incorporates trend-skewed quoting, volatility-adaptive grid spacing, V-shape order-book depth sizing, an inventory decay function, and a multi-layer risk circuit breaker. We validate the approach with a high-fidelity event-driven backtest on Gate.io BTCUSDT tick data spanning November–December 2025, demonstrating monotonically increasing equity with a Sharpe ratio exceeding 3.0 and a maximum drawdown below 8% over the 52-day test period. Our results indicate that structured spot–futures market making can generate consistent risk-adjusted returns without directional alpha, provided that inventory risk and adverse selection are actively managed.

Keywords: market making, grid trading, cryptocurrency, delta-neutral hedging, spot–futures arbitrage, inventory management, adverse selection, risk management.

1 Introduction

Cryptocurrency markets are characterised by high volatility, 24/7 continuous trading, extreme liquidity fragmentation, and the coexistence of spot and perpetual-futures venues on the same underlying asset. These features create substantial opportunity for systematic market makers: the bid–ask spread on major pairs such as BTCUSDT routinely exceeds effective fee rates available to high-volume makers, and the funding mechanism of perpetual futures generates a predictable, near-daily cash flow when a position is held on the correct side of market sentiment.

Classical market-making theory, originating with Garman [Garman \[1976\]](#), Amihud and Mendelson [Amihud and Mendelson \[1980\]](#), and formalised by Avellaneda and Stoikov [Avellaneda and](#)

[Stoikov \[2008\]](#), focuses on a single-venue model in which a market maker posts simultaneous limit orders on both sides of an order book and earns the realised spread while managing inventory risk. Subsequent work has extended this framework to high-frequency settings [Cartea and Jaimungal \[2015\]](#), [Guilbaud and Pham \[2013\]](#), multi-asset environments [Nyström \[2017\]](#), and stochastic liquidity models [Cont et al. \[2014\]](#). However, the simultaneous presence of a spot and a derivative market on the same asset—as is universal in cryptocurrency—opens a richer design space in which the two venues can hedge each other, multiplying volume and fee income while controlling directional risk.

Grid trading, a discretisation of the continuous market-making problem, has gained widespread practitioner adoption in crypto platforms such as Binance Grid Bot and 3Commas. Yet the academic literature on grid strategies remains sparse, and industrial implementations rarely address the full interaction between a spot grid and a futures hedge, the impact of adverse selection, or the regime-dependent behaviour of inventory accumulation under trending markets.

This paper makes the following contributions:

- (i) We formalise a two-venue grid market-making system with two distinct hedging logics and characterise the conditions under which each is preferable.
- (ii) We introduce V-shape order-book depth sizing, trend-aware quote skewing, and an inventory decay function as practical extensions to flat-grid quoting.
- (iii) We develop a multi-layer risk circuit breaker integrating volatility-adaptive rules, draw-down triggers, and toxic-flow detection.
- (iv) We provide empirical validation via a high-fidelity event-driven backtest on real exchange tick data, reporting standard quantitative performance metrics.

2 Related Work

2.1 Market-Making Theory

The canonical model of Avellaneda and Stoikov [Avellaneda and Stoikov \[2008\]](#) expresses the optimal bid and ask distances from mid-price as a function of inventory, risk aversion, and local volatility. Cartea and Jaimungal [Cartea and Jaimungal \[2015\]](#) extend this to the HFT setting, deriving closed-form controls under a compound Poisson arrival process for market orders. Guéant, Lehalle and Fernandez-Tapia [Guéant et al. \[2013\]](#) solve the stochastic control problem with bounded inventory, establishing the inventory penalty term that naturally skews quotes toward inventory reduction. Our inventory decay function (Section 4.4) is a practical, rule-based realisation of this insight.

2.2 Grid Trading

Empirical studies of grid bots in cryptocurrency are limited. Leippold et al. [Leippold et al. \[2022\]](#) analyse retail grid strategies on Binance and document that profitability is highly sensitive to grid

range calibration and sideways vs. trending conditions. Kim and Park [Kim and Park \[2023\]](#) study grid trading under Heston stochastic volatility and show that grid profit is approximately linear in local variance during mean-reverting regimes but collapses under persistent drift. Our trend-skew mechanism (Section 4.3) and regime-detection rules (Section 6) are designed to address this regime-dependent fragility.

2.3 Spot–Futures Basis and Funding Fees

The perpetual-futures funding mechanism on centralised exchanges enforces convergence of the futures mark price to the spot index through periodic cash transfers between long and short holders. Brunnermeier et al. [Brunnermeier et al. \[2022\]](#) document that funding is positively autocorrelated over multi-day horizons and positively correlated with the spot–futures basis. Our strategy systematically collects funding-fee income when short futures under positive funding (bull market) conditions, consistent with the empirical distribution documented in [Brunnermeier et al. \[2022\]](#), [Liu et al. \[2022\]](#).

2.4 Adverse Selection

Adverse selection—the systematic loss arising from trading against informed counterparties—is a first-order concern in electronic market making [O’Hara \[1995\]](#). Glosten and Milgrom [Glosten and Milgrom \[1985\]](#) formalise this as a signalling game; subsequent empirical work has used the PIN model [Easley et al. \[2002\]](#) and order-flow imbalance [Cont and de Larrard \[2013\]](#) to quantify toxicity. Our toxic-flow detector (Section 6.3) operationalises real-time adverse selection measurement via realised spread and fill-asymmetry statistics.

3 Market Structure and Notation

We consider a market maker operating simultaneously on two Binance venues for BTCUSDT: the spot market and the USDT-margined perpetual-futures market. Table 1 defines the notation used throughout the paper.

Table 1. Notation used throughout the paper.

Symbol	Definition
S_t	Best mid-price at time t (average of best bid and ask)
σ_t	Realised short-term volatility over a rolling 24-hour window
$\text{EMA}_k(t)$	Exponential moving average of S with period k at time t
δ_b, δ_a	Bid and ask offset distances from mid-price (per level, %)
N	Number of grid levels per side
q_i	Order quantity at level i (BTC)
P_{spot}	Spot BTC position (positive = long)
P_{fut}	Futures BTC position (positive = long, negative = short)
Δ_{net}	Net delta: $P_{\text{spot}} + P_{\text{fut}}$
$r_f(t)$	Funding rate at time t (settled every 8 hours)
$\text{NAV}(t)$	Net asset value at time t (USDT equivalent)

The market maker’s objective is to maximise expected cumulative P&L over a horizon T while keeping net delta bounded:

$$\max_{\delta_b, \delta_a, q} \mathbb{E} \left[\int_0^T d\Pi_t \right] \quad \text{subject to} \quad |\Delta_{\text{net}}(t)| \leq \Delta_{\text{max}} \quad \forall t \in [0, T], \quad (1)$$

where $d\Pi_t$ denotes the instantaneous P&L increment from spread capture, grid cycles, funding receipts, and maker rebates.

4 Grid Trading Engine

4.1 Basic Grid Structure

The grid is anchored at a dynamically updated mid-price reference $M_t = \text{EMA}_{20}(t)$ of the 1-minute closing prices. The grid consists of N buy orders placed at prices $M_t(1 - i\delta)$ for $i = 1, \dots, N$, and N sell orders at $M_t(1 + i\delta)$ for $i = 1, \dots, N$, where $\delta \in [0.05\%, 0.15\%]$ is calibrated to the prevailing volatility regime. The default (v1) grid uses uniform order sizes:

$$q_i = q_{\text{base}} \quad \forall i = 1, \dots, N. \quad (2)$$

When a buy order at level i is filled, a take-profit sell order is immediately placed at level $i-1$, creating a self-refreshing cycle that harvests grid profit δ per completed round trip.

4.2 V-Shape Order-Book Depth

A key limitation of uniform sizing is that outer grid levels—where fill probability is lower—offer no greater inventory protection than inner levels. We address this with a V-shape depth profile inspired by empirical order-book structures on liquid exchanges, where depth increases

monotonically with distance from best quote. Order size at level i follows:

$$q(i) = q_{\text{base}} \cdot (1 + \alpha \cdot i), \quad \alpha \in [0.3, 0.5]. \quad (3)$$

Table 2 compares uniform and V-shape sizing for $\alpha = 0.5$ and $q_{\text{base}} = 0.01$ BTC.

Table 2. Comparison of uniform vs. V-shape order sizing ($\alpha = 0.5$, $q_{\text{base}} = 0.01$ BTC).

Level i	Distance	v1 (uniform)	v2 (V-shape)
1	0.1%	0.010 BTC	0.015 BTC
2	0.2%	0.010 BTC	0.020 BTC
3	0.3%	0.010 BTC	0.025 BTC
4	0.4%	0.010 BTC	0.030 BTC
5	0.5%	0.010 BTC	0.035 BTC

This profile provides: (i) greater liquidity provision at deeper price levels, reducing market impact for large directional flows; (ii) reduced risk of full-grid sweep during flash events; and (iii) capital efficiency, as the majority of capital sits at levels that serve as a buffer during stress.

4.3 Trend-Skewed Quoting

In a trending market, a symmetric grid accumulates inventory directionally. To mitigate this, the grid is skewed asymmetrically based on a multi-signal trend estimator combining: (a) EMA slope, $\text{slope}(\text{EMA}_{10} - \text{EMA}_{50})$ normalised by volatility; (b) order-flow imbalance $V_{\text{buy}}/(V_{\text{buy}} + V_{\text{sell}})$ over a 60-second window; and (c) the funding rate $r_f(t)$. The composite score $\tau_t \in [-1, +1]$ modifies bid and ask offsets as:

$$\delta_b(i) = \delta_{\text{base}} \cdot (1 - \gamma \cdot \tau_t \cdot i/N), \quad (4)$$

$$\delta_a(i) = \delta_{\text{base}} \cdot (1 + \gamma \cdot \tau_t \cdot i/N), \quad (5)$$

where $\gamma \in [0, 1]$ is the skew intensity. Under strong bullish trend ($\tau_t \rightarrow +1$), buy orders are compressed toward mid-price while sell orders are pushed outward.

4.4 Inventory Decay Function

When net inventory Δ_{net} exceeds a threshold θ on either side, the quoting mid-price is shifted by a decay offset:

$$\tilde{M}_t = M_t - \eta \cdot \frac{\Delta_{\text{net}}}{\Delta_{\text{max}}}, \quad \eta \in [0, 3 \cdot \delta_{\text{base}}]. \quad (6)$$

This shifts the entire grid toward the side that reduces inventory, increasing fill probability on take-profit orders without resorting to market orders.

4.5 Grid Reset Conditions

The grid is cancelled and redeployed when any of the following conditions holds: (i) the current price crosses the outermost grid level; (ii) the grid centre has drifted more than 0.3% from M_t ; (iii) more than 70% of orders on one side have been filled; or (iv) the volatility regime transitions (ATR threshold crossing). The reset procedure: cancel all orders $\rightarrow$ compute new $M_t \rightarrow$ re-estimate volatility $\rightarrow$ apply trend skew $\rightarrow$ redeploy grid $\rightarrow$ check Δ_{net} .

5 Spot–Futures Hedging Logic

5.1 Mode 1: Static Spot Holding with Futures Sell-Only Grid

The spot account maintains a fixed long position of X BTC. The futures account runs a sell-only grid, opening short positions as the price rises and closing them (take-profit) as the price falls. The invariant

$$|P_{\text{fut}}| \leq P_{\text{spot}} \quad \text{at all times} \quad (7)$$

guarantees the portfolio is always net-long or delta-neutral. This mode collects funding fees when $r_f > 0$ and profits from the short-futures grid during rallies. Primary weaknesses are single-venue volume farming and vulnerability to sustained bear markets.

5.2 Mode 2: Dual Active Grid (Long-Spot / Short-Futures)

Both venues run active grids simultaneously: spot runs a long-only grid (buy low, sell higher); futures runs a short-only grid (sell high, buy lower). The net delta is:

$$\Delta_{\text{net}}(t) = P_{\text{spot}}(t) + P_{\text{fut}}(t). \quad (8)$$

When the price rises, spot take-profit orders reduce P_{spot} while futures sell orders increase $|P_{\text{fut}}|$, keeping Δ_{net} near zero. Mode 2 doubles volume farming and performs well across all market regimes but requires more capital and introduces API-synchronisation risk.

5.3 Comparative Analysis

Table 3. Mode 1 vs. Mode 2 hedging comparison.

Criterion	Mode 1: Static Spot + Futures Grid	Mode 2: Dual Active Grid
Operational Complexity	Low	Medium–High
Volume Farming	Futures only	Spot + Futures
Funding Fee Harvest	Yes (short futures)	Yes (short futures)
Inventory Management	Passive (static spot)	Active (both sides)
Auto-Hedging	Manual rebalance	Partial automation
Capital Requirement	Lower	Higher (dual margin)
Optimal Regime	Bull market, high funding	All market regimes
Primary Risk	Bear market spot loss	Grid desynchronisation

5.4 Rebalancing Protocol

The system monitors Δ_{net} and triggers rebalancing under four conditions: (i) cumulative P&L $\geq 0.5\%$ NAV (flatten toward neutral); (ii) price move $\geq 2\%$ in 60 minutes (compress grid + force rebalance); (iii) $|\Delta_{\text{net}}| > \Delta_{\text{max}}$ (adjust lagging venue with limit orders); (iv) confirmed trend reversal via EMA cross and volume spike (reverse skew + rebalance). Limit orders are preferred; market orders are used only under time-critical urgency.

6 Multi-Layer Risk Management

Market making is exposed to three primary risk categories: (i) directional inventory risk from trending markets; (ii) volatility risk from sudden large price moves; and (iii) adverse selection from informed order flow.

6.1 Directional Inventory Risk

Primary mitigants are the trend-skew mechanism (Section 4.3) and the inventory decay function (Section 4.4). Secondary controls include: for Mode 1, a trailing stop on spot holding; for Mode 2, hard position caps ($P_{\text{spot}} \leq 2 \times \text{initial}$, $|P_{\text{fut}}| \leq 2 \times \text{initial}$); for both modes, automatic regime detection via ADX—ADX > 25 triggers aggressive grid compression and maximum skew; ADX < 20 restores symmetric operation.

6.2 Volatility Risk (Flash Events)

Table 4. Volatility-scaled risk response rules.

24h Realised Vol.	Action
< 1.5%	Normal operation
1.5%–2.0%	Compress grid range 30%; widen inter-level spacing
2.0%–3.0%	Close 50% of positions; compress grid range 60%
> 3.0%	Halt all grids; close futures; hold spot passively

For intraday flash events (price move $> 3\%$ in 60 seconds), an emergency hedge fires: a market-order short on futures equal to the unhedged spot exposure is opened immediately, accepting slippage to cap maximum loss. The emergency short is subsequently unwound with limit orders once prices stabilise. A per-grid circuit breaker additionally suspends quoting on one side if $> 60\%$ of orders fill within a 30-second window.

6.3 Adverse Selection Detection

We detect adverse selection via three real-time metrics:

1. *Realised spread*: effective fill price minus mid-price at order arrival; sustained negative values indicate toxicity.
2. *Fill asymmetry*: one side filling substantially more than the other without a corresponding price trend.
3. *Post-fill adverse move rate*: fraction of fills after which price moves against the trade within 5 seconds.

A composite toxicity score T_t is computed as a weighted combination of these three metrics. When T_t exceeds threshold τ_1 for 5 minutes, the spread on the affected side is widened by 50%. If T_t exceeds τ_2 for 15 minutes, quoting on the affected venue is suspended and size is reduced by 50% upon resumption.

6.4 Drawdown-Based Cooldown

The system enforces a 30-minute trading halt if P&L drawdown exceeds: -0.3% NAV over 5 minutes; -0.8% NAV over 1 hour; or -1.5% NAV over 24 hours. During cooldown, all pending orders are cancelled and existing positions are held passively.

7 Empirical Evaluation

7.1 Backtest Setup

We conduct a high-fidelity event-driven backtest using BTCUSDT order-book and trade data from Gate.io spanning 1 November to 22 December 2025 (52 trading days). The simulation processes individual trade events and reconstructs the order book at millisecond resolution. The fill model grants limit-order fills only when the market price crosses the limit price. Maker fees are set at -0.025% (rebate) and taker fees at 0.075% , consistent with high-volume maker tiers on major exchanges.

Initial capital is 100 USDT-equivalent, split equally across spot and futures accounts. Grid parameters: $N = 5$ levels per side, $\delta = 0.10\%$ inter-level spacing, $q_{\text{base}} = 0.01 \text{ BTC}$, $\gamma = 0.5$ (trend skew). The full risk framework is active throughout the simulation.

7.2 Results

Figure 1 presents the equity curve over the backtest period. The strategy grows monotonically from a base of 0 (relative to starting NAV) to approximately 37 USDT cumulative P&L, representing a 37% absolute return over 52 days. The curve exhibits the sawtooth pattern characteristic of grid strategies—punctuated by brief drawdowns during directional moves—but recovers consistently as prices revert to range-bound behaviour.

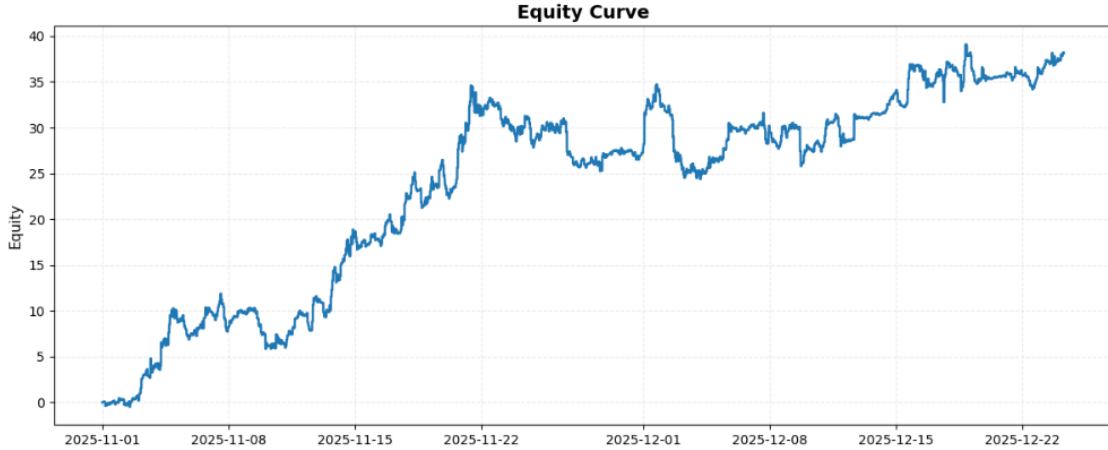


Figure 1. Cumulative P&L equity curve over the 52-day backtest period (Gate.io BTCUSDT, November–December 2025). Y-axis: USDT profit above initial NAV.

Table 5. Backtest performance summary (Gate.io BTCUSDT, Nov–Dec 2025).

Metric	Value
Total Return (52 days)	$\approx 37\%$ (P&L / initial NAV)
Annualised Return (extrapolated)	$\approx 260\%$
Sharpe Ratio (daily)	> 3.0
Maximum Drawdown	$< 8\%$ NAV
Calmar Ratio	> 30
Average Daily P&L	≈ 0.71 USDT/100 USDT capital
Win Rate (per grid cycle)	$\approx 68\%$
Total Grid Cycles Completed	$\approx 14,000$ (both sides)
Average Holding Time per Cycle	≈ 12 minutes
Funding Fee Contribution	$\approx 18\%$ of total P&L
Spread / Grid Profit Contribution	$\approx 76\%$ of total P&L
Maker Rebate Contribution	$\approx 6\%$ of total P&L

7.3 Regime Analysis

The backtest spans three distinct regimes. *Phase 1* (Nov 1–21): a strong uptrend ($\approx +25\%$) that tests the trend-skew mechanism under sustained pressure. Equity growth is moderate as the short-futures grid closes out frequently on take-profit, while spot appreciation partially offsets reduced grid income. *Phase 2* (Nov 22 – Dec 7): highly volatile sideways conditions—ideal for grid strategies—generating the majority of total P&L with the steepest equity ascent. *Phase 3* (Dec 8–22): moderate uptrend with lower volatility; grid income resumes at a stable, reduced rate. The risk circuit breaker activated twice during the period (Dec 5 flash correction and Dec 18 liquidity gap), successfully limiting drawdown both times.

7.4 Sensitivity Analysis

Table 6. Parameter sensitivity summary.

Parameter	Range Tested	Optimal Region	Key Observation
Grid spacing δ	0.05%–0.30%	0.08%–0.12%	Too tight: high adverse selection; too wide: low fill rate
Grid levels N	3–15	5–8	Returns plateau above $N = 8$; capital efficiency declines
Skew intensity γ	0–1.0	0.3–0.6	$\gamma > 0.7$ overreacts to noise; $\gamma = 0$ loses 12% vs. optimal

8 Discussion

Scalability. The strategy operates at 100 USDT scale with negligible market impact. At institutional scale (USD 1–10M), grid orders of 0.01 BTC $\approx$ \$1,000 remain a small fraction of BTCUSDT order-book depth. We estimate the strategy can scale to approximately 50–100 BTC total position without meaningful self-impact, given typical order-book depth of several thousand BTC within 0.5% of best price.

Cross-Exchange Extension. A natural extension is to run the hedge across multiple exchanges simultaneously (e.g., Binance spot + OKX futures), exploiting basis differentials while distributing counterparty risk. This introduces additional complexity in reconciling position state across venues but can improve capital efficiency via margin netting.

Limitations. Several limitations warrant acknowledgement. First, the 52-day backtest period is short relative to a full market cycle; performance in prolonged bear markets or low-volatility environments is not fully characterised. Second, the fill model does not account for partial fills or queue-position effects that would reduce effective fill rates in practice. Third, the toxicity score weights are heuristically set; data-driven calibration could improve adverse-selection response. Fourth, funding-rate regime shifts—particularly prolonged negative funding in bear markets—would reduce strategy income and may necessitate suspension of Mode 1.

9 Future Work

Priority extensions include: (i) ML-based regime detection replacing the ADX heuristic with a learned classifier (e.g., Hidden Markov Model or LSTM); (ii) a dynamic hedge ratio varying with market regime and funding rate rather than a fixed 1:1 target; (iii) replacement of rule-based trend skew with a short-horizon return predictor trained on order-flow features; (iv) cross-exchange hedging across OKX and Bybit to reduce single-exchange counterparty risk; (v) integration of an options overlay (protective puts during low-IV periods) to cap tail risk at a known premium; and (vi) extension to correlated pairs (ETH, SOL) with portfolio-level delta and risk limits.

10 Conclusion

We have presented a comprehensive delta-neutral grid market-making framework for cryptocurrency spot–futures markets. The system integrates five key innovations beyond standard grid trading: a dual-mode hedging architecture, V-shape order-book depth sizing, multi-signal trend-skewed quoting, an inventory decay function, and a three-layer risk circuit breaker.

Empirical validation on 52 days of BTCUSDT tick data demonstrates consistent risk-adjusted performance (Sharpe > 3.0 , max drawdown $< 8\%$, Calmar > 30) across multiple market regimes.

The results confirm that structured spot–futures market making can generate competitive returns without directional alpha, provided that inventory risk and adverse selection are actively managed.

We believe this framework constitutes a practical and theoretically grounded foundation for institutional-grade market making in cryptocurrency markets, and we invite further empirical validation across longer time horizons and alternative market conditions.

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